

Efficacy of clear aligner wear protocols in orthodontic tooth movement—a systematic review

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Summary

Background and objectives: Different expedited aligner wear protocols are currently in practice. This review was undertaken to systematically appraise the available evidence on the comparative efficacy of orthodontic tooth movement (OTM) across the different wear protocols.

Search methods: Two assessors conducted comprehensive searches of electronic databases, including MEDLINE (via PubMed), Scopus, Embase, Web of Science, Google Scholar, Directory of Open Access Journals, Cochrane Library, OpenGrey, and Clinical Trial Registry, till 18 February 2024. Titles and abstracts were independently screened.

Selection criteria: Prospective or retrospective studies comparing expedited wear protocols with the conventional 14-day protocol were included.

Data collection and analysis: A pre-piloted data extraction form was used. Risk of bias (RoB) assessment employed the Cochrane RoB 2.0 tool for randomized-controlled trials (RCTs) and the Newcastle–Ottawa scale for non-RCTs. The quality of evidence was evaluated using the Grading of Recommendations, Assessment, Development, and Evaluation Pro tool.

Results: Six studies, including 3 RCTs and 3 non-RCTs, were selected from 9076 records. Four studies (two RCTs and two retrospective cohort) found no statistically significant difference ($P > 0.05$) in the OTM efficacy between expedited and conventional protocols. Two studies (one RCT and one prospective cohort) found greater efficacy ($P < 0.05$) with the 14-day protocol, with the RCT reporting greater efficacy for certain movements, such as maxillary posterior intrusion, maxillary posterior distal tipping and buccal torquing, and mandibular posterior intrusion and extrusion. One RCT reported statistically insignificant difference ($P > 0.05$) in pain perception between the 10-day and 14-day protocols. Two studies demonstrated low RoB, two moderate, and two high RoB. The evidence level was very-low for OTM efficacy and high for pain perception. Meta-analysis was precluded due to significant heterogeneity among the studies.

Conclusions: Within the limitations of the study, the 7-day, 10-day, and 14-day protocols did not show any significant difference in OTM efficacy, except for certain movements that exhibited superior outcomes with the 14-day wear. Hence, a ‘hybrid aligner-wear protocol’, based on clinical judgement, might serve a better alternative in complex situations.

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Keywords: aligner; removable orthodontic appliance; wear protocol; Invisalign; efficacy; activation schedule

Introduction

Clear aligners have ushered in a transformative era in orthodontics, providing a convenient and aesthetically pleasing alternative to conventional appliances. The surge in their popularity is evident, driven by their efficacy in addressing a spectrum of orthodontic concerns, ranging from minor malocclusions to more intricate cases [1, 2]. Positioned as a favoured choice among patients, aligners not only provide an inconspicuous alternative but also afford the flexibility to maintain optimal oral hygiene [3].

Orthodontic treatment duration plays a pivotal role in determining patient satisfaction [4]. Patients typically favour treatments of shorter duration and minimal discomfort [4]. Optimizing treatment duration while ensuring effective orthodontic outcomes remains a paramount concern. Studies have demonstrated that mild to moderate malocclusions typically require 6–18 months of uninterrupted aligner wear

[5]. Conversely, complex cases necessitate extended treatment durations, spanning up to 24 months or beyond [6]. Furthermore, inter-individual variations might contribute to the heterogeneity observed in treatment durations.

The Align technology in 2016 recommended changing each set of aligners every 7 days, deviating from the traditional 14-day wear protocol [7]. This recommendation stemmed from the company’s clinical analysis involving over 200 Invisalign® patients, revealing no discernible difference in the predictability of tooth movements between the 7-day and 14-day protocols.

Few clinical trials have investigated the comparative efficacy of tooth movements with the shortened 7- or 10-day schedule as opposed to the conventional 2-week protocol [8, 9]. Notably, no systematic reviews have been published to date in this regard. Hence, the present systematic review was undertaken to compare the efficacy of orthodontic

tooth movements (OTM) between the conventional 14-day protocol and the current expedited activation schedules of clear aligners and provide a qualitative synthesis, facilitating more informed clinical decision-making.

Material and methods

Protocol and registration

The conduct and reporting of this systematic review adhered to the updated Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [10]. The review protocol was duly registered in the International Prospective Register of Systematic Reviews (CRD42021288179).

Eligibility criteria

The following population, intervention, comparison, outcomes, and study design (PICOS) criteria were used to assess studies for inclusion in this review:

Population

Individuals over 18 years undergoing clear aligner therapy (CAT) in both dental arches.

Intervention

Subjects undergoing any type of orthodontic correction, using clear aligners from any manufacturer, with an activation schedule spanning less than 14 days.

Comparison

Intervention using the same appliance with an activation schedule of 14 days.

Outcomes

Primary outcome

Efficacy of OTM assessed using occlusal indices or digital superimposition.

Secondary outcome

Pain perception assessed by grading scales or questionnaires.

Study designs

Randomized controlled trials (RCTs), prospective, and retrospective studies.

In vitro studies, preliminary reports, opinions, case reports or series, narrative reviews, and studies that involved the use of auxiliary appliances or any means of accelerating OTM were excluded.

Information sources and search strategy

Electronic databases, including MEDLINE (via PubMed), Scopus, Embase, Web of Science, Google Scholar, Directory of Open Access Journals, and the Cochrane Library, were systematically searched without imposing constraints on language and publication dates. The US National Library of Medicine database (ClinicalTrials.gov) was screened for the ongoing and unpublished trials. A search for grey literature was carried out through OpenGrey. Additionally, a manual search of the citations in pertinent studies was conducted. The initial search was conducted in October 2023 and subsequently updated until 18 February 2024. **Supplementary Appendix A** illustrates the search strategy implemented in PubMed, which was subsequently customized for other databases.

Study selection

Two evaluators (P.E. and J.M.) independently performed the screening of titles and/or abstracts in the initial phase, excluding duplicates and materials deemed irrelevant. This process was facilitated using the Rayyan software (www.rayyan.ai). Full texts were examined to reach a conclusion when titles and abstracts lacked essential details. Subsequently, the two assessors independently reviewed the full texts of studies identified in the initial screening, selectively including those that aligned with the predefined eligibility criteria. Any disparities regarding the inclusion of articles were resolved through in-depth deliberations.

Data extraction

A pre-piloted data extraction form, encompassing the following particulars, was developed: title, publication year, registration, study characteristics including design and setting, sample attributes (sample size, mean age, gender, type of malocclusion), randomization, blinding, number of aligner trays, activation schedule, treatment time, observation period, primary and secondary outcome measures, assessment method, statistical analysis, and results. It was performed by one of the assessors (J.M.) and cross-checked by the other (E.P.).

Risk of Bias (RoB) within studies

The revised Cochrane RoB tool was employed for assessing the RCTs and the Newcastle–Ottawa scale for the prospective and retrospective cohort studies [11, 12]. It was independently carried out by two assessors (P.E. and J.M.) and subsequently duplicated to mitigate potential bias. The RCTs were assessed under five domains namely, randomization process, deviations from intended interventions, measurement of outcome, missing outcome data, and selection of reported result [11]. The Newcastle–Ottawa Scale involved three domains, namely, participant selection with four items, comparability (one item), and outcome including three items [12]. For each non-RCT, a maximum of one star is allocated for each item in participant selection and outcome domains and a maximum of two stars for comparability.

Data synthesis

A qualitative synthesis of the data extracted from the included studies is provided. A quantitative analysis of the findings was intended only when there was substantial homogeneity among the studies.

Certainty of evidence

The appraisal of the quality of evidence for clinical recommendation utilized the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) Pro tool [13]. The assessment was carried out independently by two authors (P.E. and J.M.). The following domains were taken into account: number of included studies, study design, RoB, imprecision, inconsistency, indirectness, and other considerations including publication bias. For each outcome, the evaluation of evidence levels was performed individually for RCTs and non-RCTs, and the higher level of certainty was considered for each outcome.

Results

Study selection and characteristics

A total of 9076 records were identified through literature search. Removal of duplicate records and application of

eligibility criteria led to the exclusion of 7459 results. An additional 1583 records were removed after initial screening, leaving 34 records for full-text evaluation. Subsequent scrutiny led to exclusion of 28 articles, and the reasons for exclusion are presented in [Supplementary Appendix B](#). In conclusion, six studies were included in the present review, encompassing three RCTs [8, 9, 14, 15], one prospective cohort study [16], and two retrospective cohort studies [17, 18]. Notably, one of the RCTs comprised a three-part article [14, 15, 19], with the first two parts being included in this review and considered as a single study due to the involvement of the same samples [14, 15]. The third part was excluded since it did not address the specific aims under investigation [19]. Two of the included studies were registered in the Clinical Trial Registry platform [8, 16]. The PRISMA flow diagram is presented in [Fig. 1](#).

The comparative efficacy of OTM between different wear protocols was the sole objective in two RCTs [8, 9], while it was the primary aim in one RCT [14, 15] and the secondary objective in others [16–18]. Four studies examined a variety of tooth movements [8, 9, 15, 16], whereas two specifically focussed on individual tooth movements, such as the labio-palatal movement of maxillary central incisors [18] and rotation of mandibular canines [17]. All the studies included a

control group following a 2-week activation schedule, which was compared against the shortened 1-week protocol in three studies [14, 15, 17, 18] and 10-day protocol in one RCT [8]. The remaining two studies compared 1-week as well as 10-day protocols with the standard [9, 16].

All the studies included subjects over 18 years old and treated using Invisalign® only, involving both arches. The sample size across the six studies totalled 449, with the exclusion of 54 samples from the final analyses. This comprised five samples from an RCT due to insufficient data [9], nine samples from another due to reduced wear time (< 22 hours—one subject, non-compliance with the prescribed protocol—three subjects, inadequate time to collect necessary records—five subjects) [8], four samples (one from each of the four groups) from a retrospective study as the teeth in question moved in a direction opposite to that planned [18], and with only 15 out of 51 patients completing treatment in the trial by Bollen *et al.* [14] for reasons such as improper appliance fit—23, rebooting—10, and patient unwilling to adhere to the prescribed protocol—3. The study by Castroflorio *et al.* [16] did not specify the sample sizes within each group.

All studies, except one [14], employed the commercial Invisalign® material for aligner fabrication. Bollen *et al.* [14] used two distinct stiffness variants of the Invisalign®

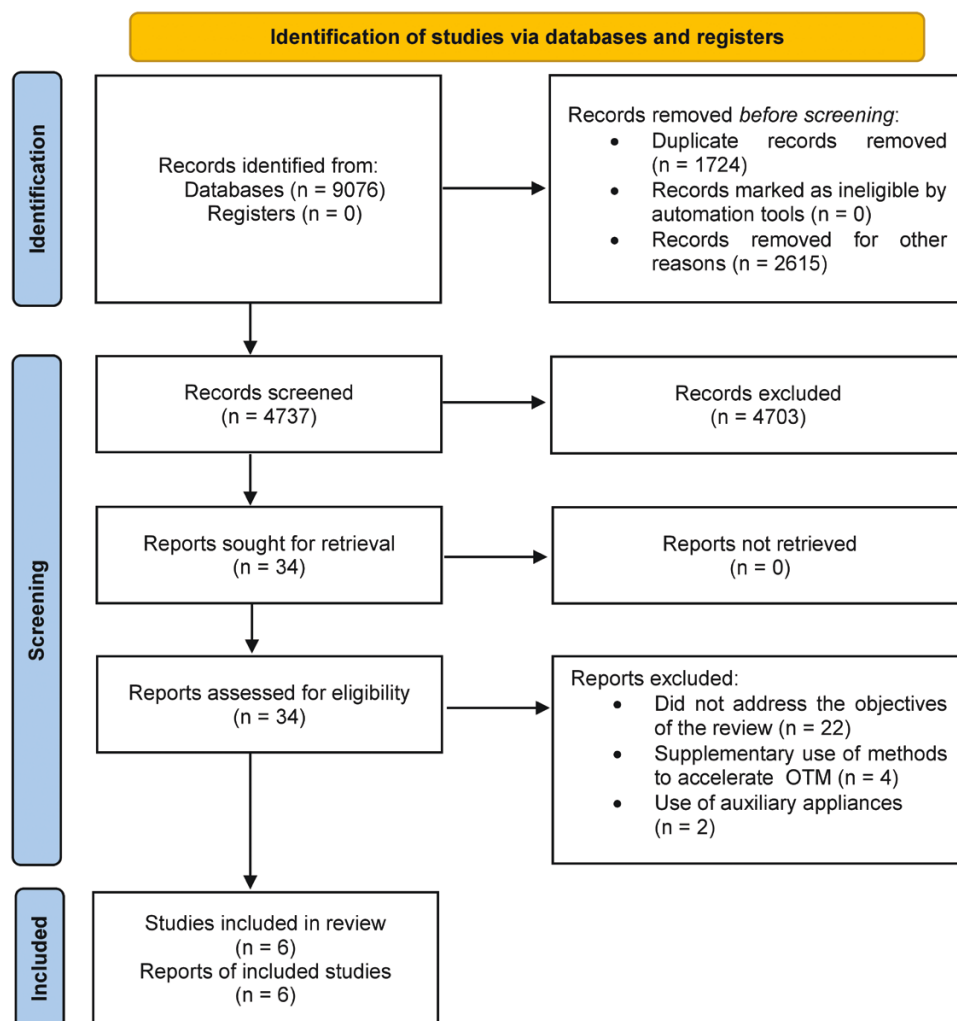


Figure 1. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram

Table 1. Summary of study characteristics.

Study	Design	Groups/ activation schedule	Sample size (M/F)	Mean age (SD)	No. of aligner sets Mx/Md	Mean treatment time Mean (SD)	Follow-up period	Primary outcome measure	Secondary outcome measure(s)	Results/Outcome
Al-Nadawi <i>et al.</i> [9] 2021	RCT	G1: 7-day G2: 10-day G3: 14-day	G1: 27 (7/20) G2: 25 (12/13) (2 excluded) G3: 23 (11/12) (3 excluded) Gender- $P < 0.05$	G1: 36.3y G2: 34.3y G3: 35.4y $P > 0.05$	G1: 21/19 G2: 22/20 G2: 20/20 $P > 0.05$	G1: 5m G2: 8m G3: 9m $P < 0.05$	Start of treatment till the completion of initial series of aligners	Best-fit 3D superimposition of predicted and achieved outcomes using eModel® 9.0 Compare software. Differences between predicted and achieved tooth positions were calculated.	-	Mx- G3 showed significantly ($P < 0.05$) greater accuracy compared to G1 for posterior TM such as intrusion (0.3 vs. 0.4 mm), distal crown tip (2.8° vs. 3.9°), buccal crown torque 2.8° vs. 4.4°) Md- G3 showed significantly ($P < 0.05$) greater accuracy compared to G1&2 for posterior intrusion (0.3 vs. 0.4 mm) and extrusion (0.2 vs. 0.3 vs. 0.3 mm).
Zhao <i>et al.</i> [8]. 2023	RCT	G1: 10-day G2: 14-day	G1: 33 (8/25) (3 excluded) G2: 30 (6/24) (6 excluded) $(P > 0.05)$	G1: $\leq 45y-32$ >45y- 1 G2: $\leq 45y-28$ >45y- 2 $(P > 0.05)$	G1: ≥ 16 sets	G1: 160 days G2: 224 days (calculated from stage 1 to stage 16.	Start of treatment till the completion of 16 sets of aligners. Records were collected at three time points. T0- before initial insertion T1- end of stage 8 T2- end of stage 16.	Two methods: 1. Stable structure method using unmoved teeth for superimposition. 2. Shell-to shell deviation method (whole surface best-fit method). The initial, actual, and predicted models were superimposed using Geomagic® Studio software.	Pain perception assessed using Visual Analog Scale (VAS) on first, second, third, and seventh day for each aligner set.	OTM efficacy was similar in both groups for all types of movements at T1 and T2 ($P > 0.09$). VAS scores were also similar at all time points assessed in all groups ($P > 0.05$).

Table 1. Continued

Study	Design	Groups/ activation schedule	Sample size (M/F)	Mean age (SD)	No. of aligner sets Mx/Md	Mean treatment time Mean (SD)	Follow-up period	Primary outcome measure	Secondary outcome measure(s)	Results/Outcome
Bollen <i>et al</i> [14], 2003	RCT	G1: soft plastic/ 7-day G2: hard plastic/ 7-day G3: soft plastic/ 14-day G2: hard plastic/ 14-day	G1: 12 (6/6) G2: 12 (3/9) G3: 14 (3/11) G4: 13 (3/10) Gender- (P = 0.36)	G1: 34.9y (11.6) G2: 33.5y (8.6) G3: 36.9y (11.4) G4: 31.0y (12.4) (P = 0.58)	G1: 31(15.3)/28.4(14.9) G2: 33.8(9.6)/27.9(7.6) G3: 28.8(11.5)/30.0(11.5) G4: 27.3(8.0)/27.8(7.1) Mx: P = 0.52 Md: P = 0.95	NA	Start of treatment till the completion of initial aligner series.	-	Ability to complete treatment was assessed by the rate of completion of initial series of aligners.	Subjects following the 14-day protocol (37%) showed greater tendency to complete treatment compared to 7-day subjects (21%), although not statistically significant (P < 0.2). No statistically significant difference in completion rates between hard and soft appliances (P < 0.69).
Clements <i>et al</i> [15], 2003	Same as Bollen <i>et al.</i> 2003						Start of treatment till the completion of initial aligner series or until it was determined the aligner was not fitting well (study end point)	Weighted PAR score and average incisor irregularity (AII) index were used to measure the pre-treatment and end-point models. Amount of space closure and closure percentage at the end point for each extraction space were calculated.	-	No statistically significant difference in weighted PAR scores (P = 0.2786) as well as maxillary (P = 0.79) and mandibular AII (P = 0.5603) scores between 7-day and 14-day activation schedules or hard and soft materials.

Table 1. Continued

Study	Design	Groups/ activation schedule	Sample size (M/F)	Mean age (SD)	No. of aligner sets Mx/Md	Mean treatment time Mean (SD)	Follow-up period	Primary outcome measure	Secondary outcome measure(s)	Results/Outcome
Castroflorio <i>et al.</i> [16], 2022	Prospective cohort	G1: 7-day G2: 10-day G3: 14-day	79 (23/56)	30.8y (12.0)	27 (15)/ 25 (11)	9.8m (3.8)	Till the completion of treatment.	Landmark-based superimposition of predicted and achieved outcomes using Geomagic® Qualify software. Superimposition accuracy was improved by the best-fit alignment.	-	The 14-day protocol (-0.12) reduced the lack of correction by 12%, which was significantly greater ($P = 0.009$) than the 7-day protocol (0.21). The difference between the 7-day and 10-day (-0.09) groups were not statistically significant.
Stephens <i>et al.</i> [17], 2022	Retro-spective cohort	G1: optimized rotation attachments/7-day G2: optimized rotation attachments/14-day G3: conventional rectangular attachments/14-day	G1: 25 G2: 25 G3: 25	NA (included patients over 18y)	NA	NA	Till the completion of treatment. Records were collected at three time points. T1- Initial T2- Predicted finish T3- Achieved finish	The initial, predicted, and actual models were independently analysed using Geomagic® Control X software.	-	The efficiency of mandibular canine rotation in G1 (81.5%) was not significantly different ($P > 0.05$) from G2 (76.5%). The median achieved change was very similar among the three groups: G1: 14.6° G2: 14.8° G3: 15.2°
Wei <i>et al.</i> [18], 2023	Retro-spective cohort	G1: Labial/ 7-day G2: Palatal/ 7-day G3: Labial/ 14-day G4: Palatal/ 14-day	G1: 26 (after exclusion of 1 sample) G2: 18 (after exclusion of 1 sample) G3: 24 (after exclusion of 1 sample) G4: 20 (after exclusion of 1 sample)	NA (included patients over 18y)	NA	NA	Till the completion of treatment. Records were collected at three time points. T1- Initial T2- Predicted finish T3- Achieved finish	Best-fit 3D superimposition of predicted and achieved outcomes using Geomagic® Control X software.	-	No statistically significant difference ($P > 0.05$) between the two wear protocols for the direction of maxillary central incisor movements as well as the centre of rotation deviation.

F, female; G, group; M, male; m, months; Mx, maxilla; Md, mandible; NA, not available; RCT, randomized-controlled trial; SD, standard deviation; y, year.

material: the hard variant, which exhibited double the stiffness of the commercial material, and the soft, characterized by one-tenth the stiffness of the commercial material. The daily wear time was recorded using a smartphone application by Zhao *et al.* [8], with a mean recorded time of 22.38 ± 0.61 h/day.

Four studies incorporated non-extraction cases [9, 16–18], whereas one RCT included both extraction and non-extraction subjects [14]. Additionally, another trial specifically addressed cases necessitating substantial OTM, involving more than 16 sets of aligners [8]. The RCTs enrolled subjects with Class I, II, and III malocclusions [8, 9, 14, 15], while the prospective study included Class I and mild Class II cases with mild–moderate crowding or spacing [16]. Of the retrospective studies, one exclusively focussed on Class I cases [18], with the malocclusion status not specified in the other [17].

The follow-up period in the RCTs was till the completion of the initial series of aligners [8, 9, 14, 15]. Conversely, it extended until the end of treatment in the non-RCTs [16–18]. The characteristics of the included studies are detailed in Table 1.

RoB within studies

Of the three RCTs, one exhibited an overall low RoB [8] and two showed high RoB [9, 14, 15]. The principal factors contributing to high RoB were the absence of information on allocation concealment, lack of appropriate analysis to estimate the effects of intervention assignment, and no details on sample size estimation. Additionally, Al-Nadawi *et al.* [9] did not report the inter- and intra-rater reliability assessment, while the RCT by Bollen *et al.* [14, 15] exhibited substantial missing data, potentially impacting the study outcomes.

One of the non-RCTs showed a low RoB [16], while the other two showed a moderate risk [17, 18]. There was a lack of information to ensure comparability of the study groups across all the three studies. Two studies were further downgraded due to the presence of the outcome at the start of the study [17, 18]. In addition, Wei *et al.* [18] did not report blinding of the assessors. The RoB evaluation in the RCTs is presented in Fig. 2 (generated using Risk-of-bias Visualization tool) [20] and non-RCTs in Table 2.

Data synthesis

A meta-analysis was precluded by substantial heterogeneity in the methodology and result reporting across the included studies. Consequently, our approach was restricted to a descriptive synthesis of results.

Results of individual studies

Primary outcome

Four of the six studies reported no statistically significant difference ($P > 0.05$) in the efficacy of OTM among the different activation schedules tested [8, 15, 17, 18]. This included two RCTs [8, 15] and two retrospective studies [17, 18]. Three among them made a comparative assessment between the 7-day and 14-day schedule [15, 17, 18], while one compared between 10 and 14 days [8]. The method of assessment in two of the four studies was 3D superimposition of the predicted and actual models using the Geomagic® software [8, 18]; while the initial, predicted, and actual models were independently analysed using the same software in one study [17]. In contrast, Clements *et al.* [15] used occlusal indices such as the Peer Assessment Rating index (PAR) and the Average Incisor Irregularity (AII) index to score the pre-treatment and end-stage study models.

Al-Nadawi *et al.* [9] and Castroflorio *et al.* [16] studied the 7-day, 10-day, and 14-day wear protocols and found a statistically significant difference between them, with greater efficiency in the OTM with the 14-day protocol. Both used software-based 3D superimposition of the predicted and actual models. Among the 12 types of OTM assessed by Al-Nadawi *et al.* [9], significantly ($P < 0.05$) greater accuracy was reported for three posterior movements (intrusion, distal tip, and buccal torque) in the maxilla compared to the 7-day wear and two posterior tooth movements (intrusion and extrusion) in the mandible compared to the 7-day and 10-day wear. All comparisons in the maxilla between the 10-day and 14-day wear yielded insignificant results. Nevertheless, no significant differences were observed for all anterior tooth movements in either arch.

Castroflorio *et al.* [16] studied the predictability of OTM in all three planes of space with Invisalign and found a 12% reduction in the lack of correction with the 14-day protocol compared to the 7-day wear. However, the difference between the 7-day and 10-day wear was statistically insignificant.

Secondary outcomes

Zhao *et al.* [8] further studied the variations in pain perception associated with different wear protocols, employing the visual analog scale (VAS). The scores were recorded after the insertion of each new pair of aligners on days 1, 2, 3, and 7 from stage 1 to stage 16. Both the study groups (10-day and 14-day) exhibited comparable VAS scores at all the examined time points. The scores were highest on day one and at stage one in both groups.

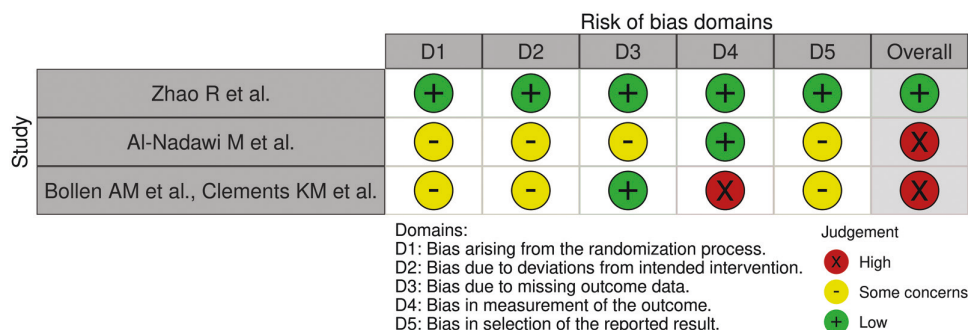


Figure 2. Summary of risk of bias in the randomized controlled trials.

Table 2. Risk of bias assessment in non-randomized cohort studies using the Newcastle–Ottawa scale [12].

Study	Representativeness	Selection of the unexposed	Ascertainment of exposure	Presence of outcome	Comparability	Assessment of outcome	Follow-up	Adequacy of follow-up	★
Castroflorio <i>et al.</i> [16].	★ (b)	★ (a)	★ (a)	★ (a)	○	★ (a)	★ (a)	★ (a)	7
Stephens <i>et al.</i> [17].	★ (b)	★ (a)	★ (a)	○ (b)	○	★ (a)	★ (a)	★ (a)	6
Wei <i>et al.</i> [18].	★ (b)	★ (a)	★ (a)	○ (b)	○	○ (d)	★ (a)	★ (b)	5

Stars are added for each parameter based on the criteria provided in the Newcastle–Ottawa scale; Circle represents not reported/ undertaken.

The ability of the patients to complete treatment following different activation schedules was assessed by Bollen *et al.* [14]. There was no statistically significant difference between the 7-day and 14-day subjects, although the 14-day subjects showed a greater tendency to complete treatment.

Certainty of evidence

The GRADE evidence profile for the efficacy of OTM indicated a ‘very low’ certainty for the 7-day versus 14-day as well as 10-day versus 14-day comparisons (Table 3). The comparison between 10-day and 14-day protocol for the pain perception showed a ‘high’ level of evidence (Table 3).

Discussion

Orthodontic treatment duration is a critical aspect of orthodontic care, impacting patient compliance, treatment completion rate, and overall oral health. The chosen orthodontic appliance or modality also influences treatment duration. Studies have shown CAT is associated with reduced treatment duration compared to conventional fixed appliances [6, 21].

In an attempt to further expedite treatment, Align Technology introduced the shortened activation schedule, deviating from the conventional 14-day wear [7]. While the company’s internal analysis indicates no significant differences in treatment efficacy between the 7-day and 14-day activation schedules, existing literature presents a mixed picture. Hence, the present systematic review was undertaken to synthesize the available evidence on the comparative efficacy of OTM with various wear protocols.

Summary of evidence

The 7-day and 10-day activation schedules were tested against the conventional 14-day wear. Two studies assessed both protocols [9, 16], while others tested either one against the standard [8, 14, 15, 17, 18].

Al-Nadawi *et al.* [9] reported the 14-day protocol to have significantly greater accuracy in producing maxillary posterior intrusion, maxillary posterior distal tipping and buccal torquing, compared to the 7-day protocol. Conversely, no significant difference was noted between the 10-day and 14-day wear for all maxillary tooth movements, which is in line with the results of Zhao *et al.* [8]. In the mandible, the 14-day wear was associated with significantly greater accuracy in posterior intrusion and extrusion, compared to 7-day and 10-day wear. This contrasts with the results of Zhao *et al.* [8], who reported no significant difference between the 10-day and 14-day wear.

Al-Nadawi *et al.* [9] found all the anterior movements in either arch to show no significant difference between

the different activation schedules. This corroborated the findings of both retrospective cohort studies, which found insignificant differences in the efficiency of mandibular canine rotation and labio-palatal movement of maxillary central incisors between the 7-day and 14-day protocols [17, 18].

Castroflorio *et al.* [16] compared the predicted and achieved outcomes to evaluate the predictability of OTM with aligners. They found a lack of correction for all tooth movements assessed, with the discrepancy being 12% less pronounced with the 14-day protocol compared to the 7-day.

Four of the six studies reported no significant difference in the OTM efficacy between the expedited activation schedules and the conventional [8, 15, 17, 18]. This finding may be attributed to the temporal dynamics of inflammatory mediators associated with OTM. Research indicates that the peak levels of these mediators occur within two days of force application and return to baseline within one week [22].

Furthermore, Kapoor *et al.* [23] have elucidated the release patterns of bone resorption markers, such as interleukin-1 beta, interleukin-8, tumour necrosis factor, and receptor activator of nuclear factor kappa ligand (RANKL), following force application. These markers were found to release within an hour of force application, reaching peak levels within 24 h. Subsequently, a decline in their concentrations is observed at 2 days, 14 days, and 21 days post-application. These temporal dynamics suggest a potential mechanism underlying the observed consistency in OTM efficacy across different activation schedules.

The difference in pain perception was assessed by one RCT, wherein no statistically significant difference between the 10-day and 14-day groups was reported [8]. Notably, the pain was maximum on day one and with the initial pair of aligners in both groups. Hence, it may be prudent to plan comparatively lesser tooth movements in stage one to enhance patient compliance.

Given the little difference in OTM efficacy and pain perception between the different activation schedules, the adverse effects, if any, of such rapid pace of force application should also be investigated. The expedited activation schedules prompt concerns regarding the increased risk of root resorption or compromised periodontal health as they could potentially surpass the physiological limits of the periodontal tissues. Conversely, longer wear periods may ensure optimal force application to the teeth, fostering adequate remodelling of the periodontium. This might result in more stable and sustainable orthodontic outcomes, potentially minimizing the likelihood of relapse. Nonetheless, research aimed at addressing these concerns remain outstanding.

Table 3. Grading of Recommendations, Assessment, Development and Evaluation (GRADE) evidence profile.

Certainty assessment		No. of patients					Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	[intervention]	[comparison]	Relative (95% CI)	Absolute (95% CI)
Efficacy of OTM with 7-day vs. 14-day wear protocol										
2	Randomized trials	Serious ^a	Very serious ^b	Not serious	Not serious	None	51	52	-	-
2	Non-randomized studies	Serious ^c	Not serious	Not serious	Not serious	None	71	69	-	-
1	Non-randomized studies	not serious	Not serious	Serious ^d	Not serious	None	NA	NA	-	-
Efficacy of OTM with 10-day vs. 14-day wear protocol										
2	Randomized trials	serious ^a	Very serious ^e	Not serious	Not serious	None	58	53		
Pain perception with 10-day vs. 14-day wear protocol										
1	Randomized trials	Not serious	Not serious	Not serious	Not serious	None	30	24		

CI, confidence interval; GRADE, Grading of Recommendations, Assessment, Development and Evaluation; NA, not available; OTM, orthodontic tooth movement; RoB, Risk of Bias.

^aBased on RoB assessment using the Cochrane RoB tool.

^bMethod of assessment is different in both RCTs and the results are contradictory. Al-Nadawi *et al.* reported greater efficacy with 14-day protocol for certain tooth movements, while Clements *et al.* reported no significant difference.

^cBased on RoB assessment using the Newcastle–Ottawa scale.

^dEvidence obtained is more restrictive.

^eResults are contradictory. Al-Nadawi *et al.* reported greater efficacy with 14-day protocol for certain tooth movements, while Zhao *et al.* reported no significant difference.

Understanding the biological and biomechanical intricacies underlying different OTM is essential for devising the most appropriate wear protocol tailored to individual patients. The findings of Al-Nadawi *et al.* [9] underscore the nuanced efficacy of the 14-day protocol for certain types of tooth movements. Hence, a customized ‘hybrid aligner-wear protocol’ integrating the 7-day wear regimen for simpler tooth movements and 14-day wear for more complex ones may be strategically planned.

Clinical implications

The choice between a 7-day, 10-day, or 14-day activation schedule might depend on various factors, including the complexity of the case, patient comfort, and the orthodontist’s treatment philosophy. More intricate tooth movements might benefit from the stability provided by a 14-day schedule, while simpler cases might progress well with a 7-day or 10-day activation schedule. Rather than employing a one-size-fits-all approach, the adoption of a tailored ‘hybrid aligner-wear protocol’ might offer a promising alternative.

One might anticipate a reduction in the number of aligner trays with the expedited protocols, but this is not the case in reality. However, the ‘hybrid aligner-wear protocol’ can potentially reduce the number of aligner trays per case, mitigating the environmental burden [24].

Strengths, limitations, and future direction

The research protocol was officially registered in the PROSPERO open database, and an extensive search across numerous databases was undertaken. Adherence to the PRISMA guidelines was ensured during the conduct and reporting phases.

Substantial methodological heterogeneity among the included studies precluded a quantitative synthesis. Two studies exhibited a high RoB, lowering the certainty of evidence.

More robust clinical trials focussing on complex malocclusions are required. Any potential adverse effects associated with expedited activation schedules, such as root resorption, need to be explored.

Conclusions

A summary of the findings is as follows:

1. Four studies, including two RCTs, reported no significant difference in the efficacy of OTM across the different activation schedules.
2. Two studies, including one RCT, demonstrated greater accuracy in OTM with the 14-day activation schedule. The RCT reported greater accuracy in producing maxillary posterior intrusion, maxillary posterior distal tipping and buccal torquing, and mandibular posterior intrusion and extrusion.
3. Results from a single RCT reported no difference in pain perception between the 10-day and 14-day activation schedules, with the most pain recorded on day one and with the initial pair of aligners.
4. The evidence level ranged from ‘very-low’ for OTM efficacy to ‘high’ for pain perception.

Within the constraints of the study, no significant differences in the effectiveness of OTM were observed among the 7-day, 10-day, and 14-day protocols, except for certain

tooth movements that demonstrated better results with the 14-day regimen. The proposed ‘hybrid aligner-wear protocol’ may offer a more nuanced approach, potentially optimizing treatment outcomes across a spectrum of clinical scenarios.

Conflicts of interest

None declared.

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Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Supplementary material

Supplementary material is available at *European Journal of Orthodontics* online.

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